From Plausibility Machines to Narrative Minds

Vision, Attention, and the Architecture of Artificial Cognition

Your Name

Your Institution

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Outline

An Analogy: Human and Computer Vision

- **Edge detection** — low-level feature extraction

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Why this analogy

Raw input $\rightarrow$ abstraction $\rightarrow$ recognition is a useful template for thinking about cognitive architecture generally. Where does narrative fit in this pipeline?

An Analogy: Human and Computer Vision

- **Edge detection** — low-level feature extraction
- **Abstraction** — features aggregate into intermediate representations
- **Object recognition** — abstraction resolves into a labeled category

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Plausibility Machines

- Chiang and Tufekci: LLMs as plausibility machines (Chiang 2024; Tufekci 2026) — producing text that reads as coherent without underlying understanding
- The move from *simulating* consciousness to *having* consciousness is often treated as small, or even inevitable

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Claim

That leap is in fact massive, and unlikely.

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- **What it is:** procedural circuits — comparable to functions in a CS sense
- **What is (partly) missing:** valence and salience

“Attention Is All You Need” (Vaswani et al. 2017)

- What attention consists of, for an LLM
- Stacked transformers and attention
- Loss minimization and static associative valuation

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- Webb and Graziano's **Attention Schema Theory** (Webb and Graziano 2015): awareness is *an internal model of attention*

"A creature with an attention schema..."

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Open possibility

Could an explicit, internal model *of the system's own attention* — not just attention weights themselves — point toward AI architectures with a closer semblance of awareness? (cf. Section "What Architecture Is Closer?")

Coglinks and Uncertainty

- Circuits imitated
- Predefined decision choices

Time, Prediction, and Memory

- Predictive processing vs. uncertainty
- Episodic and semantic memory

Phenomenological Experience and Emotion

Open question

What would have to be true of a system for phenomenological experience and emotion — not merely their outward signatures — to be present?

Narrative in Cognition (Miller 2024)

- As **active inference**

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- As **active inference**
- As an **abstraction scaffold**

What Architecture Is Closer?

- Self-salience models
- Less terrifying possibilities that simulate emotion

What Training Is Required?

- Direction 1: [To be developed]
- Direction 2: [To be developed]

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A closer semblance of self.

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2. Predictive processing (with respect to dynamic inputs)?
3. Nested experiential hierarchies — including reflection on what experiences reveal about self-function?

Thus

A closer semblance of self.

Notes / Parking Lot I

- **Oh no Waymo —**
  possible example of imitation-without-understanding in the wild

  Waymos and other AIs harnessed to bodies perform processing operations related to moving through the world with particular sets of prescribed motivations. There's lots to question about the notion of 'motivations', which I hope to do in another forum. But focusing for now just on the problem of regulating a body moving in time through space, detecting environmental conditions is an ongoing, dynamic problem. At its core, it requires recognizing other agents and dynamic forces in the world. That is, slow feature analysis is a baseline for experiential processing (Song and Zhao 2022). It builds temporal parsing into experiencing. Slow feature analysis combines with valuative processing (salience and valence) that is itself dynamic and interactive. In terms of attention, a stabilization in valuation can be a catalyst of shifting attention, which marks temporal experience with subjective meaning.

  Open questions: Can an experiencing AI have a semblance of these processes? How might it relate to the question of whether the AI has subjective experience?

- **Hebbian learning vs. gradient descent —**
  Hebbian as local; gradient descent as global

Thank You

Thank you!

Questions?

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